

WORKSHOP REPORT

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23BTT29

Workshop on Protective and Smart Textiles in Indutech
PSG College of Technology, Coimbatore
Centre of Excellence – Indutech

1. Introduction

A workshop on Protective and Smart Textiles in Indutech was conducted at the Centre of Excellence (CoE) for Industrial Textiles at PSG College of Technology. The Centre focuses on research and development in industrial textiles and promotes innovation in technical textile applications.

The workshop aimed to provide both theoretical knowledge and practical exposure to students in the field of protective and industrial textiles.

2. Overview of Technical Textiles

Technical textiles are materials designed primarily for functional purposes rather than aesthetic appeal. These textiles are used across various sectors such as agriculture, construction, automotive, medical, and industrial applications.

The focus is on performance characteristics like strength, durability, water repellency, filtration, insulation, and flame resistance.

3. Industrial Textile Products

Industrial textiles include coated and laminated fabrics, filtration materials, insulation textiles, and specialty products used in advanced applications such as aerospace and automotive industries.

Coated and laminated textiles enhance durability and resistance properties, making them suitable for harsh environments.

4. Nonwoven Textiles

Nonwoven textiles are produced without traditional weaving or knitting processes. Methods such as needle punching, spunlace, melt-blown, and spunbond are used.

These materials are widely used in filtration, hygiene products, medical textiles, and insulation.

5. Machinery and Technology

Various machines such as hot melt coating machines, nonwoven production machines, and UV flatbed printing machines were introduced during the session.

These machines demonstrate the integration of mechanical and material engineering in textile manufacturing.

6. Testing and Evaluation Methods

Testing plays a crucial role in ensuring quality and performance of textiles. Mechanical testing includes tensile strength and bursting strength tests.

Surface testing involves abrasion resistance and water repellency tests. Analytical testing includes UV weathering and microscopic analysis.

7. Flame Retardant Textiles

Flame retardant textiles are designed to resist ignition and slow down fire spread. Mechanisms include heat absorption, char formation, and gas phase inhibition.

Various chemicals such as aluminium hydroxide and phosphorus compounds are used for flame retardant treatments.

8. Thermal Conductivity

Thermal conductivity determines heat transfer through textile materials. It depends on fiber type, thickness, porosity, and moisture content.

Heat transfer occurs through conduction, convection, and radiation.

9. Lab Visit

The second session included a lab visit where students observed machinery operation and testing procedures.

This provided practical exposure to industrial textile processes and quality control methods.

10. Conclusion

The workshop provided valuable insights into protective and smart textiles. It helped in understanding industrial applications, testing methods, and advanced textile technologies.

Overall, it was an informative and enriching experience.

11. Images





SURFACE CHARACTERIZATION TESTING LAB

PROCESS TENSIOMETER



DESCRIPTION: To determine wear resistance of surface coatings
APPLICATION: Textiles, Ceramic and other materials
STANDARDS: ASTM D2480, D 3450, D 4215, D 4828 & ISO 11997.

WET SCRUB ABRASION TESTER



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MARTINDALE ABRASION TESTER



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TABER ABRASION TESTER



DESCRIPTION: To determine abrasion resistance of materials
APPLICATION: Textiles, Coatings and other materials
STANDARDS: ASTM D3399, ASTM D4065 and more.



ANALYTICAL TESTING LAB

FSEM WITH EDX



DESCRIPTION: To determine surface morphology & elemental analysis
APPLICATION: Textiles, Composites & other materials

WIDE ANGLE XRAY DIFFRACTION



DESCRIPTION: To determine phase identification of a crystalline material
APPLICATION: All crystalline materials

SIMULTANEOUS DSC & TGA



DESCRIPTION: To characterize the thermal properties of a polymer
APPLICATION: Organic and inorganic materials

UV SPECTROSCOPY



DESCRIPTION: For determination of different analytes, such as transition metal ions, highly conjugated organic compounds and biological macromolecules
APPLICATION: All appropriate materials

FTIR



DESCRIPTION: To obtain an infra spectrum of absorption, emission, photo conductivity of a solid, liquid or gas. (Chemical group identification)
APPLICATION: All polymeric materials

UV - WEATHEROMETER



DESCRIPTION: To study the accelerated life testing of materials under UV exposure
APPLICATION: Appropriate materials
STANDARDS: AATCC TM26, AATCC TM26-4, ASTM D1442, ASTM D3424 and many more.

UV TRANSMITTANCE ANALYSER



DESCRIPTION: To determine UV resistance of a material
APPLICATION: Textiles and other materials
STANDARDS: AG/NZS 4399.1/PM6, EN 18758-1/2/3/4, AATCC 30/31/32

HYDRO HEAD TESTER



DESCRIPTION: To determine water flow resistance of a material
APPLICATION: Plastics, Textiles and other materials
STANDARDS: AATCC 127, ASTM F 1870, BS 2582, DIN 53865, EN 17754, EN 20061, FZ/T 01004 Buthers



FIRE TESTING LAB

VERTICAL FLAMMABILITY TESTER



DESCRIPTION: To determine vertical combustion behaviour and flame spread pattern of a given material
APPLICATION: Textiles and similar materials
TEST STANDARDS: EN ISO 6040, EN ISO 15025 etc..

HORIZONTAL FLAMMABILITY TESTER



DESCRIPTION: To determine horizontal combustion behaviour and flame spread pattern of a given material
APPLICATION: Textiles and similar materials
TEST STANDARDS: DIN 75200, ISO 8795 etc.

INCLINED FLAMMABILITY TESTER



DESCRIPTION: To determine inclined combustion behaviour at 45 degree of a given material
APPLICATION: Textiles and similar materials
TEST STANDARDS: ASTM D 1230, BS 2469 etc..

LIMITED OXYGEN INDEX



DESCRIPTION: To determine minimum concentration of oxygen that will support flaming combustion of Textiles in a glowing mixture of oxygen and nitrogen
APPLICATION: Textiles and similar materials
TEST STANDARDS: ASTM D-2868